PRACTICAL-2

Observe and list the functions of front panel and back panel ports of a PC.

A. Objective:

To observe and list the functions of front panel and back panel ports of a Personal Computer (PC).

B. Expected Program Outcomes (POs)

PO1: Apply knowledge of engineering fundamentals to identify the front panel and back panel ports of a PC and explain their functions.

PO5: Apply appropriate hardware resources and engineering tools to observe, identify, and verify PC ports and connectors during laboratory practice.

PO8: Function effectively as an individual or as a member of a team while performing the practical.

PO9: Communicate observations by documenting the identified ports and their functions in a clear and systematic manner.

C. Expected Skills to be developed based on competency:

This practical is expected to develop the following skills:
1. Identify front and back panel ports.
2. Match peripherals with appropriate ports.
3. Explain the function of each port.

D. Expected Course Outcomes (COs)

Identify PC hardware components and explain the functions of front and back panel ports.

E. Practical Outcome (PrO)

1. Correctly identify front and rear panel ports.
2. Select the appropriate port for connecting external devices.

F. Expected Affective Domain Outcomes (ADOs)

1. Work effectively as an individual/team member.
2. Follow laboratory safety and ethical practices.

G. Prerequisite Theory:

A port is basically a physical docking point which is basically used to connect the external devices to the computer, or we can say that A port act as an interface between the computer and the external devices, e.g., we can connect hard drives, printers to the computer with the help of ports.

Features of Computer ports:

- We can connect external devices to the computer with the help of ports and cables.
- These are basically slots on motherboard where we connect external devices, or we can plug in external devices through cables.
- Mouse, keyboards, printers, speakers are some examples of external devices that connected to the computer through ports.

Components of the Front Panel
1. Power Button

- Used to switch the computer ON or OFF.
- Connected directly to the motherboard through the front panel header.
- A long press (4–5 seconds) forces the computer to shut down during system hang.

2. Reset Button

- Restarts the computer without turning off the power supply.
- Useful when the operating system becomes unresponsive.

3. Power LED

- Indicates whether the computer is powered ON.
- Usually Green, White, or Blue.

4. Hard Disk Drive (HDD/SSD) Activity LED

- Blinks whenever data is read from or written to the storage device.
- Helps indicate disk activity.

5. USB Ports

USB (Universal Serial Bus) ports connect external peripherals.

6. Audio Ports
Headphone Jack (Green)

- Audio output.
- Connects headphones or speakers.

Microphone Jack (Pink)

- Audio input.
- Connects microphones.

Memory Card Reader (Optional)

Back Panel of a Personal Computer

The back panel contains motherboard ports used for permanent connections such as monitor, keyboard, mouse, printer, network cable, and speakers.

Components of the Back Panel

1. PS/2 Port

- Used for keyboard and mouse.
- Purple → Keyboard
- Green → Mouse

2. USB Ports

Used for:

- Keyboard
- Mouse
- Printer
- Scanner
- Webcam
- External Hard Disk

3. VGA Port (Video Graphics Array)

- 15-pin analog video connector.
- Blue color.
- Connects CRT and LCD monitors.

Maximum Resolution:
1920 × 1080 (depends on hardware)

4. HDMI (High Definition Multimedia Interface) Port

Features:

- Carries digital video and audio.
- Supports Full HD, 4K and 8K displays.
- Used for monitors, TVs and projectors.

Advantages over VGA:

- Better picture quality
- Audio and video through single cable
- Supports higher resolutions

5. DisplayPort

- Digital display interface.
- Supports multiple monitors.
- Higher bandwidth than HDMI.
- Used in gaming and professional graphics workstations.

6. DVI Port (Digital Visual Interface)

- Supports digital video.
- Better quality than VGA.
- Common in older LCD monitors.

7. RJ-45 Ethernet Port

- Used for wired LAN connection.
- Supports:
  - 10 Mbps
  - 100 Mbps
  - 1 Gbps
  - 2.5 Gbps
  - 10 Gbps (advanced systems)

LED Indicators:

- Green → Link
- Orange → Data transmission

9. Serial Port (COM Port)

- 9-pin connector (DB9)
- Used in industrial controllers and networking devices.
- Mostly obsolete.

10. Parallel Port (LPT)

- 25-pin connector.
- Used for old printers and scanners.
- Rarely available in modern PCs.

H. Work Situation:

- Faculty will demonstrate different types of ports available in computer system.
- Faculty must form a group of two or three students.
- Students group will observe different ports, connectors and its connection.

I. Resources Required:

- Desktop PC

J. Procedure:

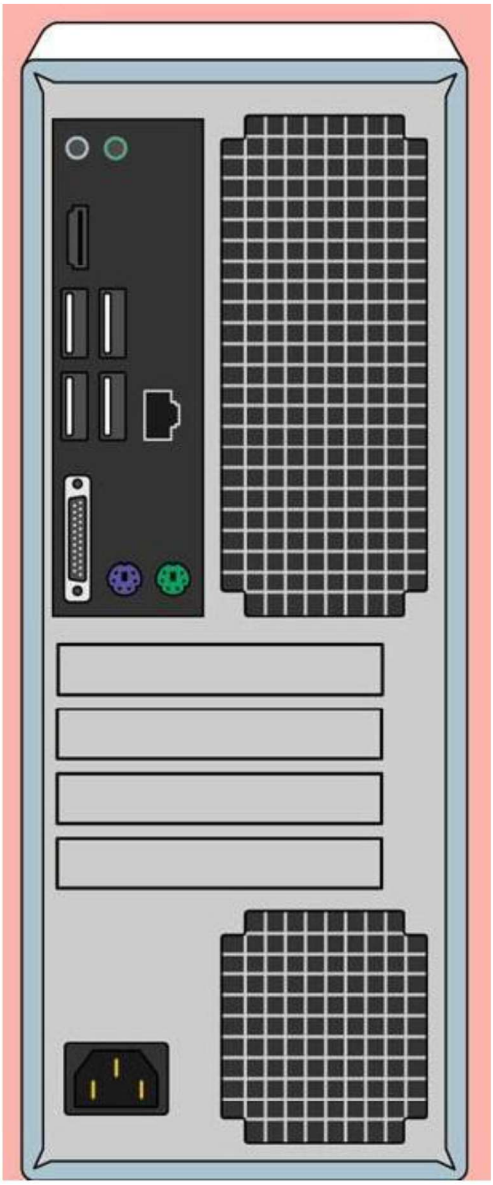
1. Switch OFF the computer and disconnect the power supply.
2. Observe the front panel and identify all buttons, LEDs, and ports.
3. Observe the back panel and identify all available ports and connectors.
4. Note the name and function of each front and back panel port.
5. Connect suitable peripherals to the identified ports (if available) and verify their purpose.
6. Record your observations in the observation table.
7. Compare the identified ports with the motherboard/manual and discuss their applications.
8. Disconnect all peripherals and restore the laboratory setup.

K. Questions:

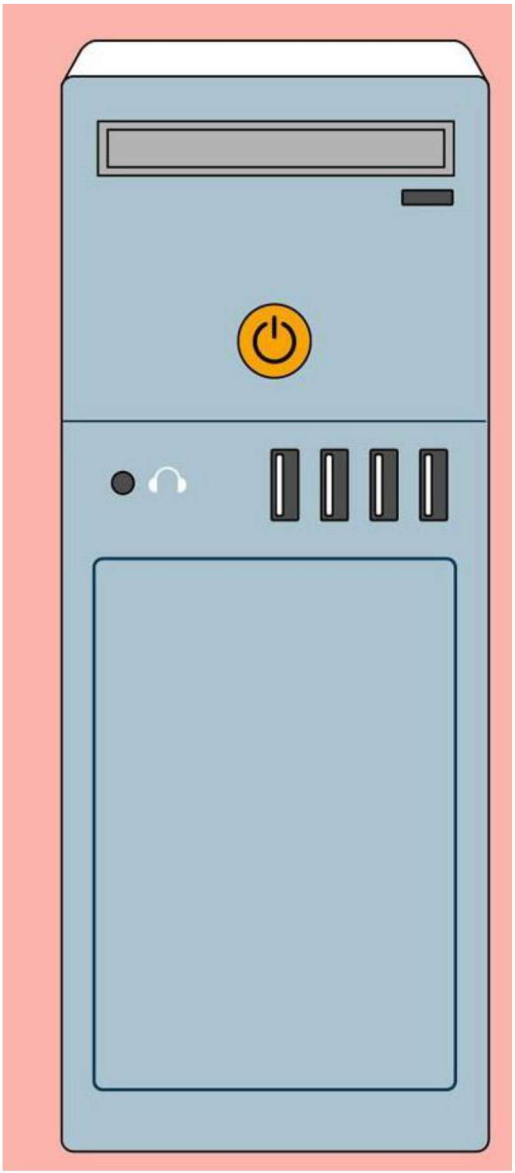
1. Different ports of PC and its use.

2. Which connector is used to connect computer to the network?

3. Label the Back side of the PC.



4. Label the frontside of the PC.



5. Differentiate between a VGA port and an HDMI port.